

Urban Form and Transport

ARCH 305 Architecture and Town Planning

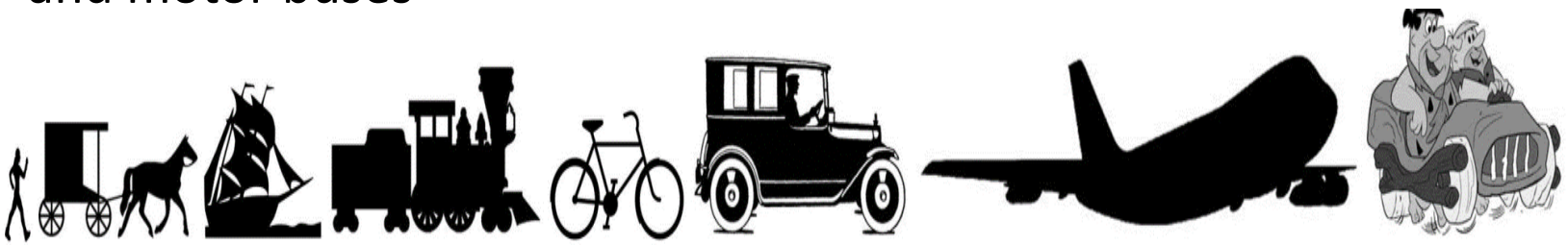
Walking Cities" to "Automobile Cities

- Cities and means of travel grow together
- Urban form is determined by transport technologies available.
- Urban transportation services define;
  - the geographic area in which people function
  - limiting how far one may travel to work
  - acquire food, exchange services, and visit friends

Walking Cities" to "Automobile Cities

The history of urban transportation is a story of the evolution of technology;

- from walking → to riding animals → to riding in groups on vehicles pulled by animals → and eventually to cars, trains, electric trains, and motor buses



- It is a story of gradually increasing **speed, vehicle capacity, and range of travel**
- The phenomenon has **shaped cities and structured** the lives of those who live in them

Industrial Revolution

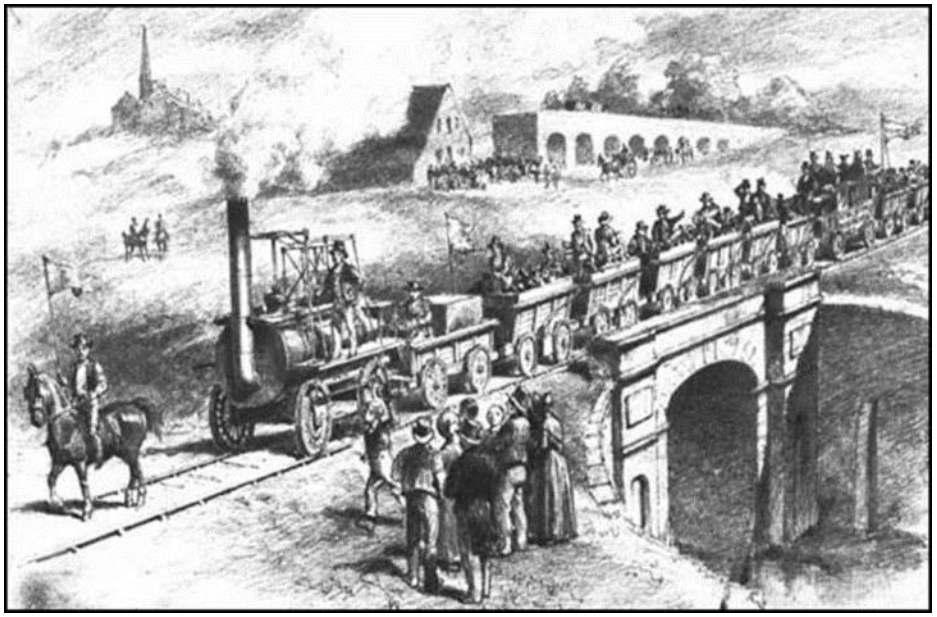
- Industrial – Having to do with industry, business or manufacturing

- Revolution – a huge change or a change in the way things are done

- Industrial Revolution – a change from making things by hand to making them in factories.

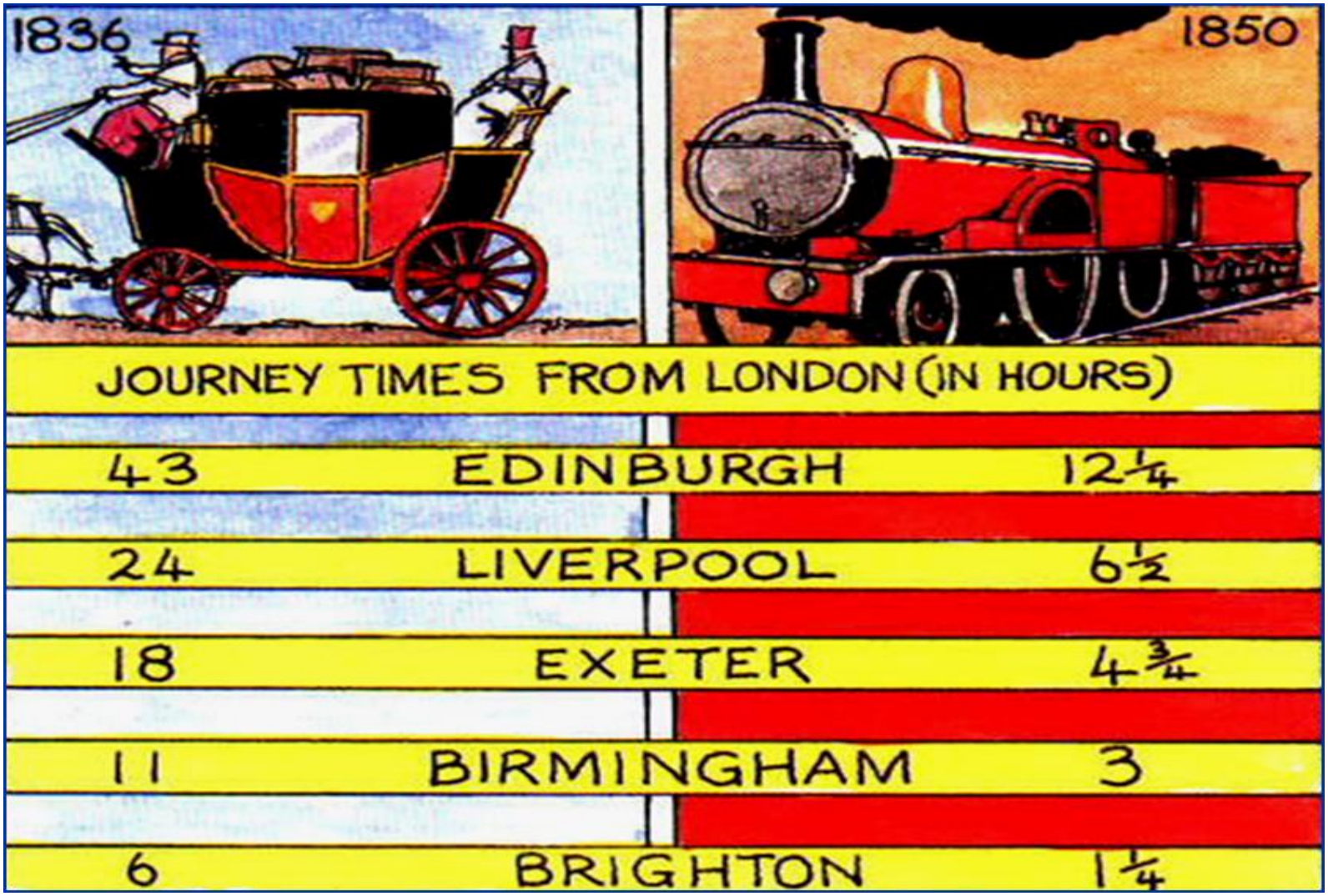
Industrial Revolution

- Before the Industrial Revolution, people relied on the horse and their own feet to get around.
- With the invention of the steam locomotive, transportation took a huge step forward.

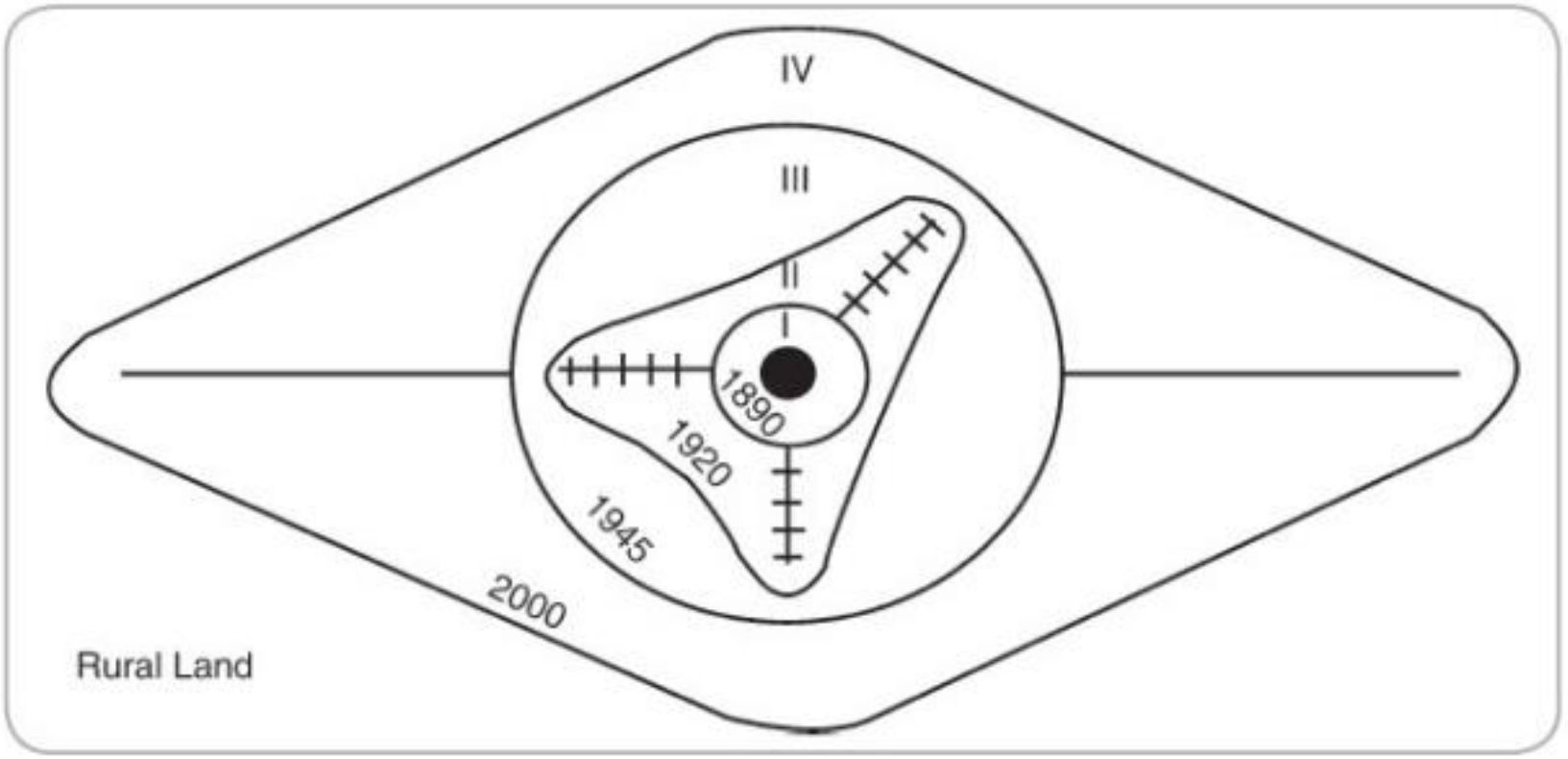




Industrial Revolution Impact



Typical City Growth and Evolution of Transport



Muller, P.O. (2004) "Transportation and Urban Form: Stages in the Spatial Evolution of the American Metropolis", in S. Hanson and G. Giuliano (eds) The Geography of Urban Transportation, 3rd Edition, New York: Guilford, pp. 59-84.


30 mins

I. Walking-Horsecar Era (1800–1890)

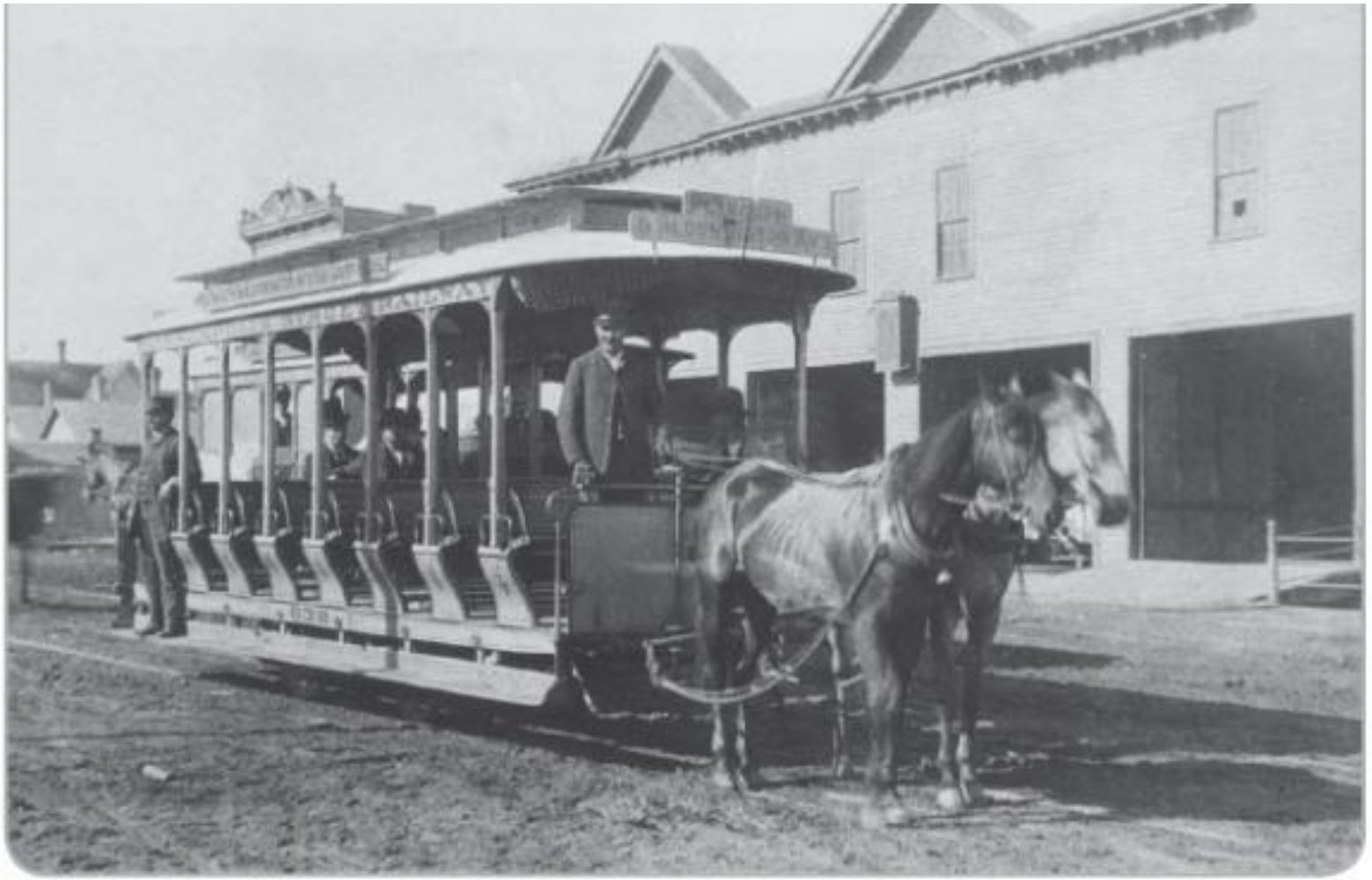
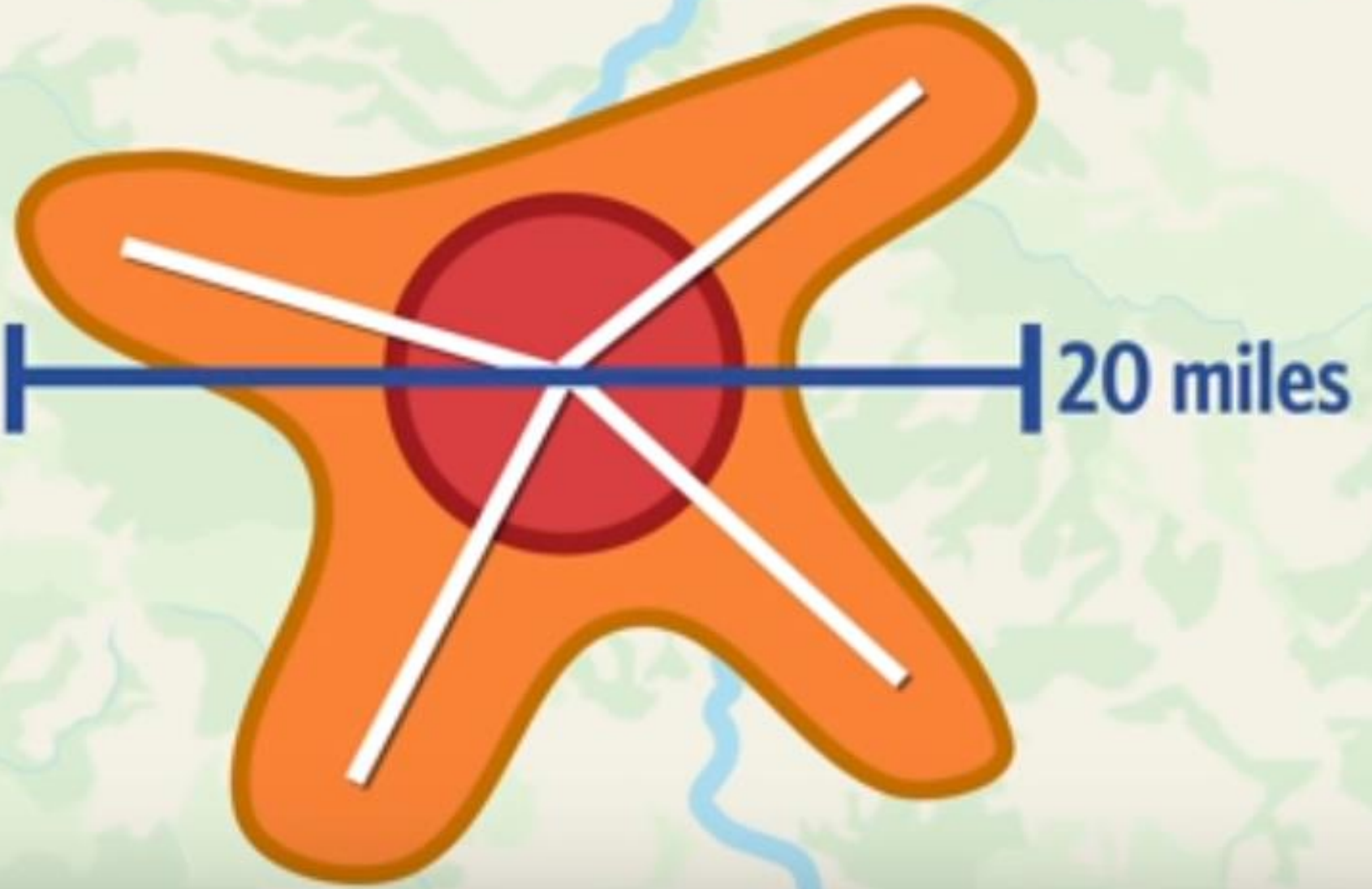


FIGURE 3.6. The horsecar introduced mass transportation to the teeming American city. This dictograph was taken in downtown Minneapolis in the late 1880s. Source: Minnesota Historical Society.

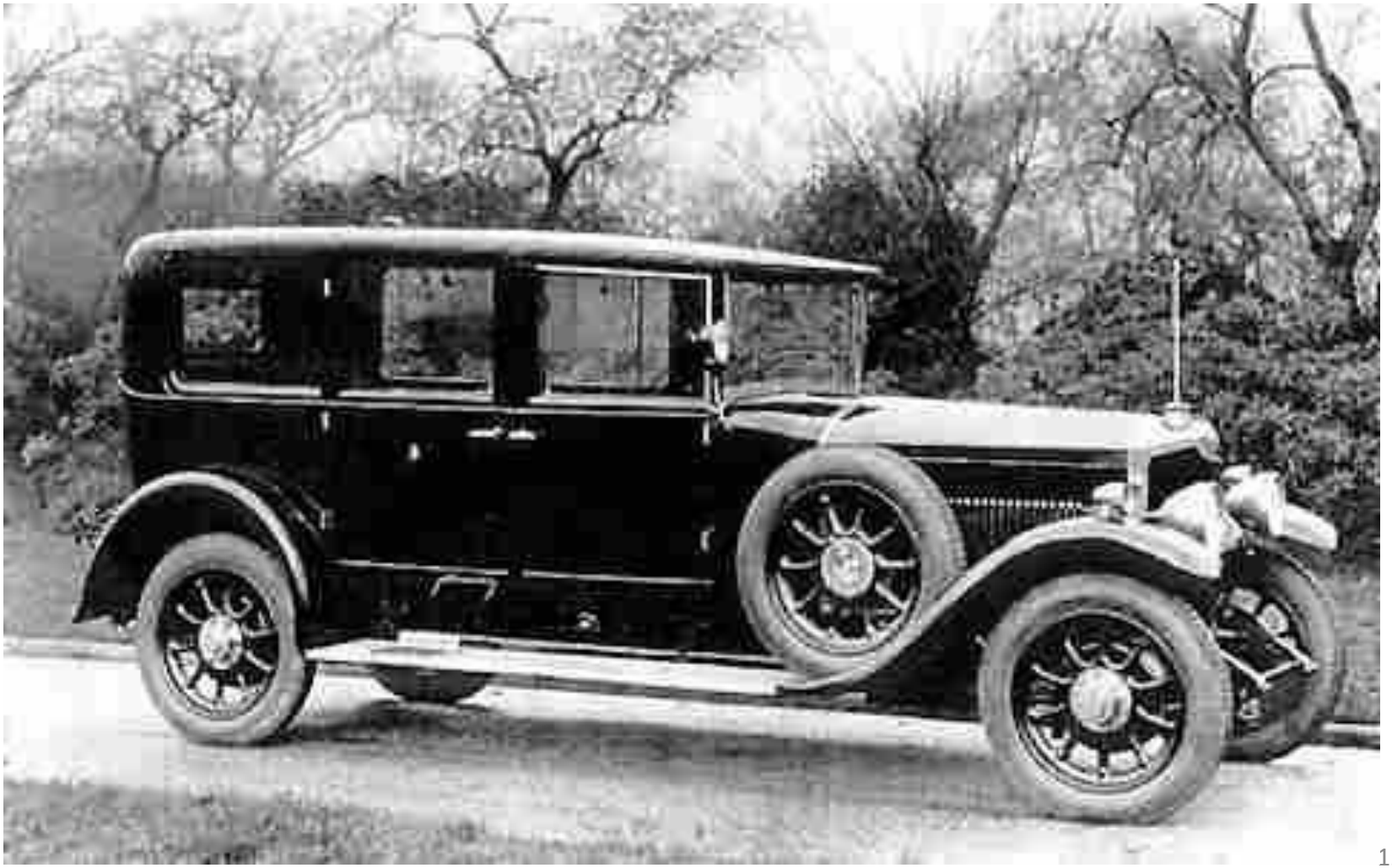
II. The Electric Streetcar Era (1890–1920)



FIGURE 3.10. The automobile did not become a major force in the central city until the post–World War II era, but its presence can already be detected by the 1930s. This photograph was taken in St. Paul in 1932. Source: Minnesota Historical Society.


20 miles

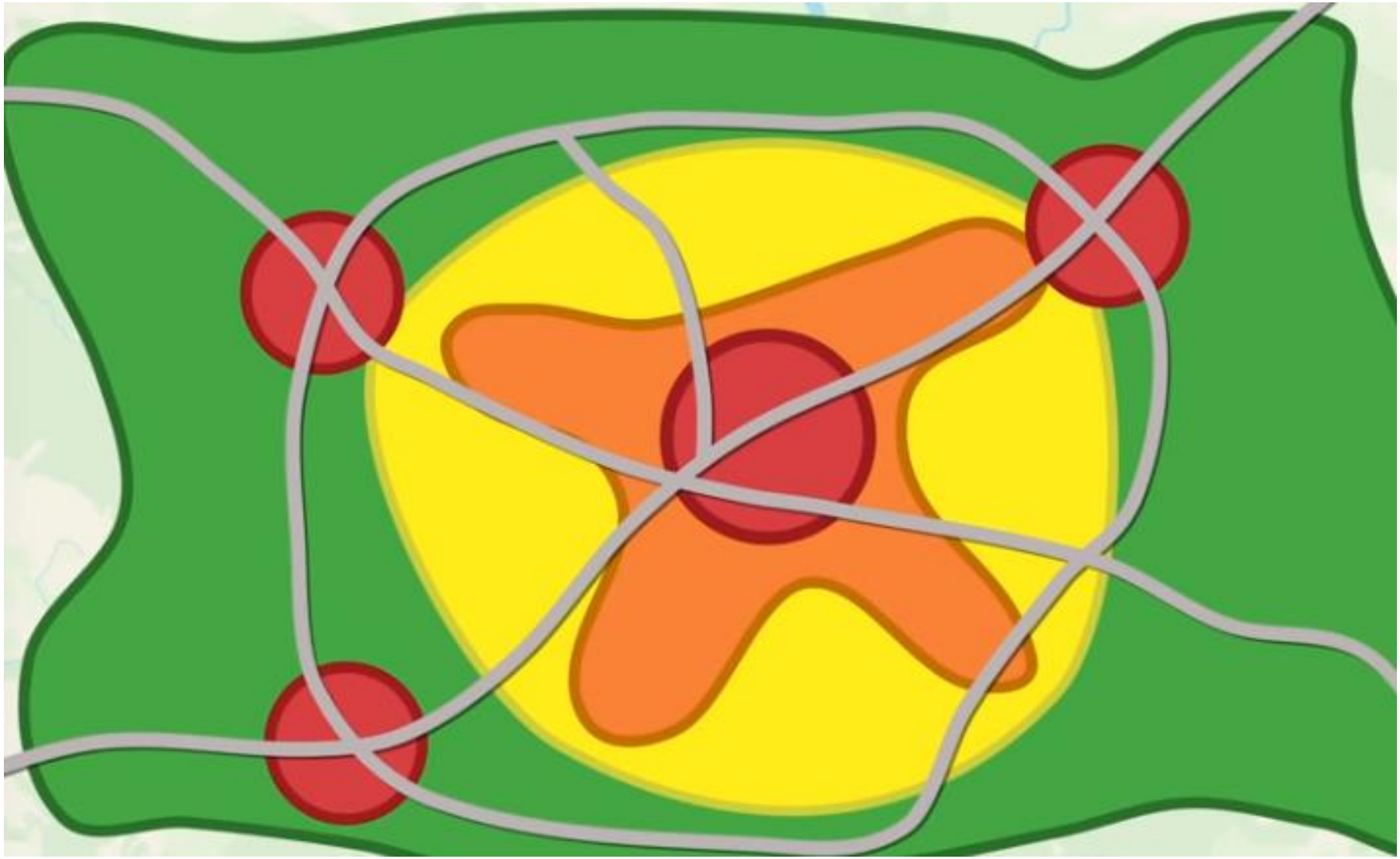
III. The Recreational Automobile Era (1920–1945)

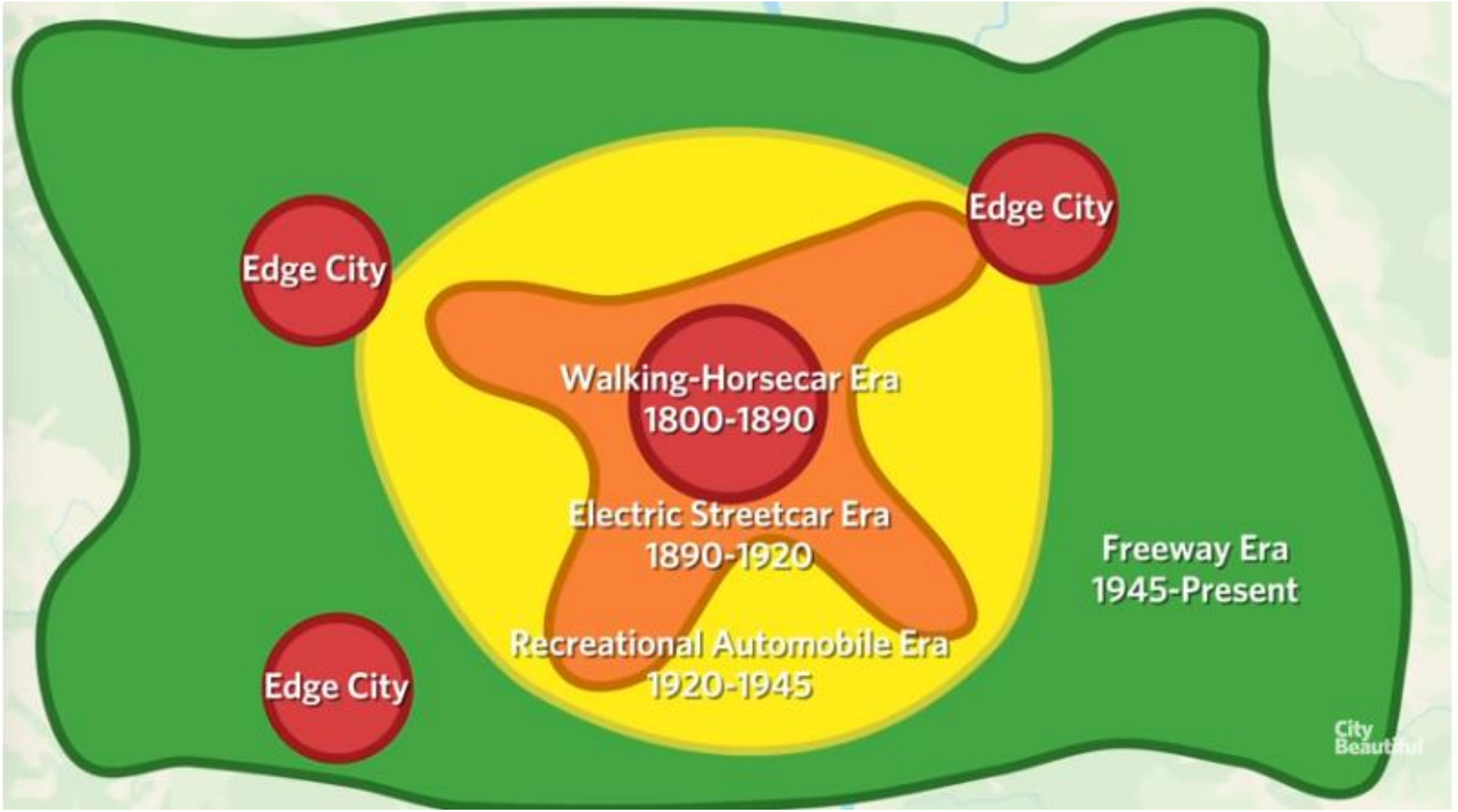




iv. **Freeway Era (1945–Present)**







Source: How do cities grow?
https://www.youtube.com/watch?v=HO0BZqQ42Ic

One Hour Commuting According to Different Urban Transportation Modes

Marchetti's constant

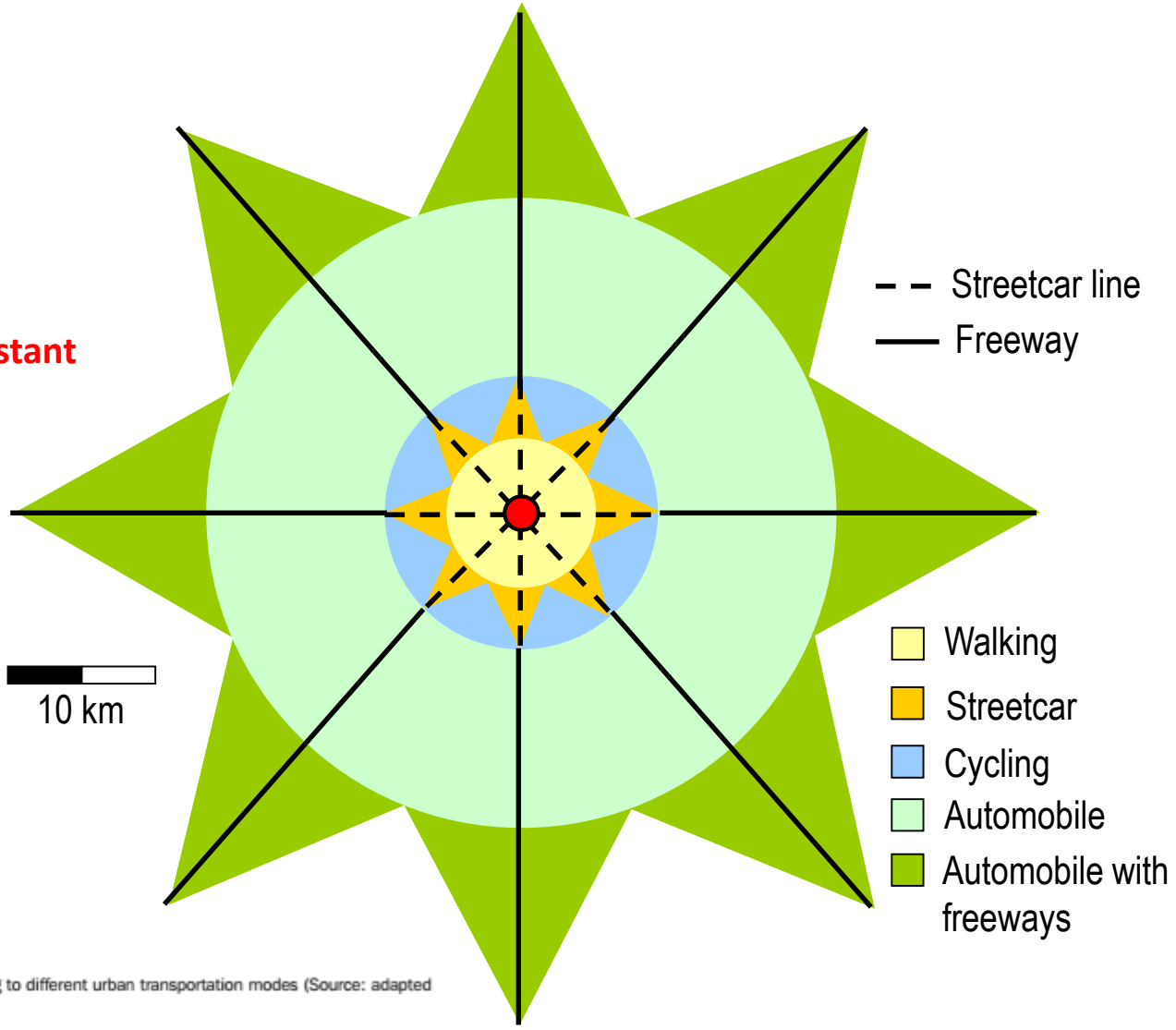
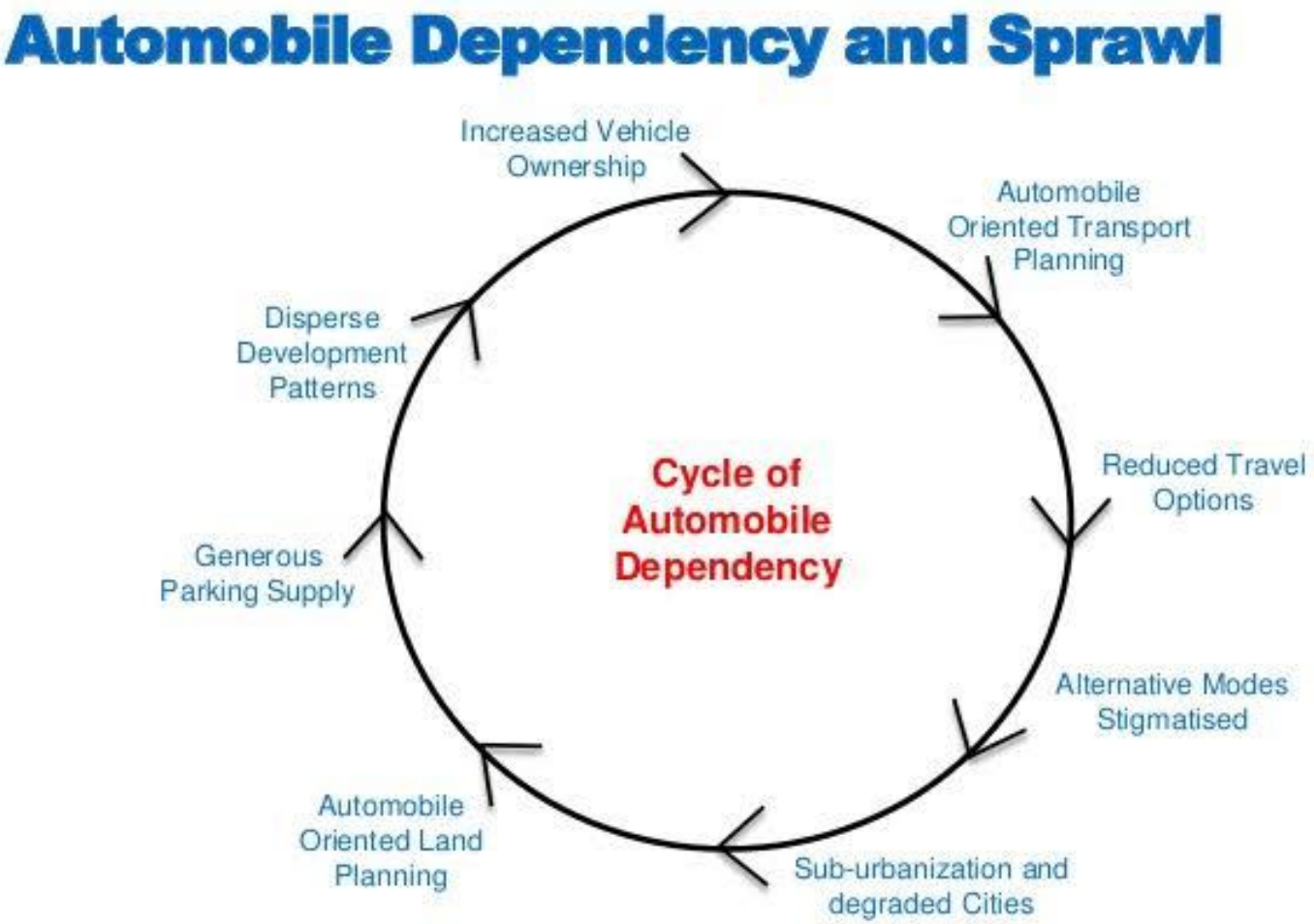


Figure 7.2 One-hour commuting according to different urban transportation modes (Source: adapted from Hugill, 1993, p. 213)

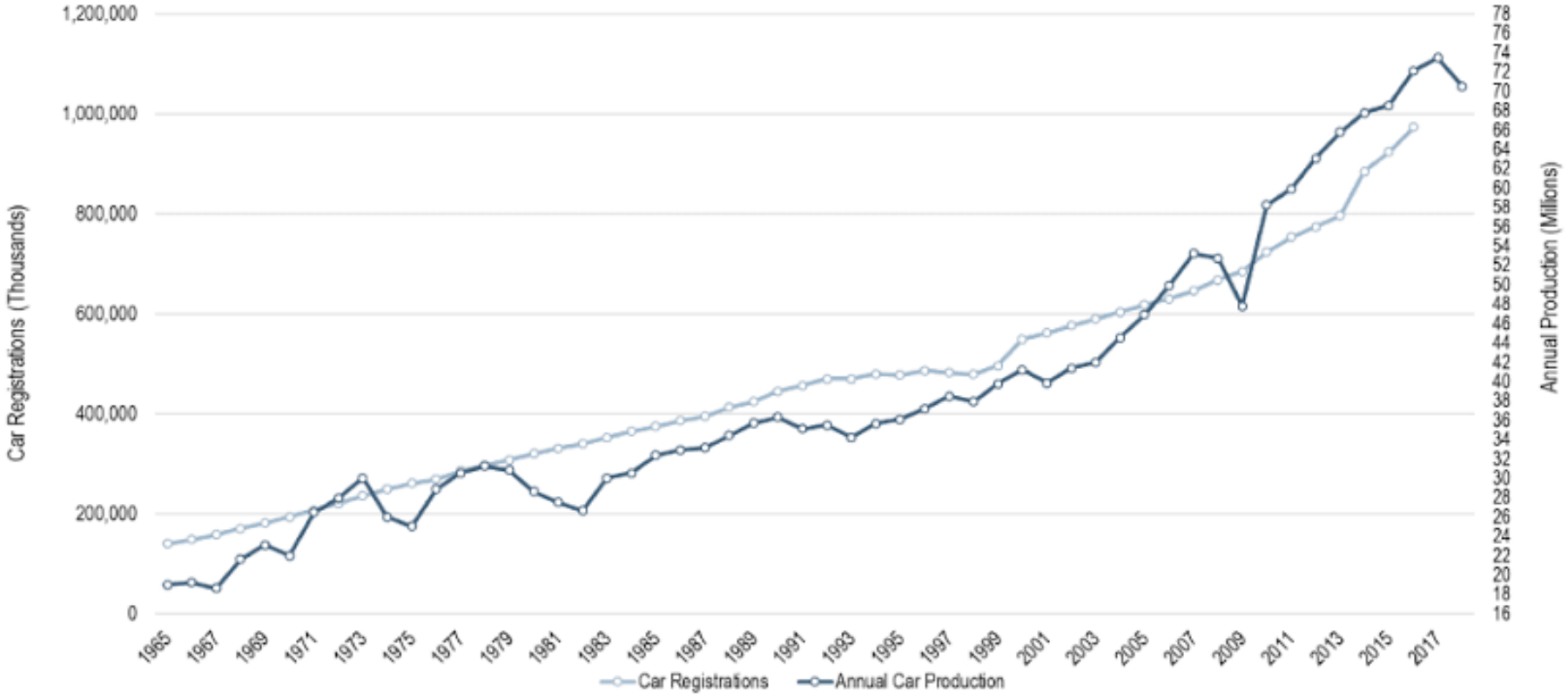
Automobile Dependency



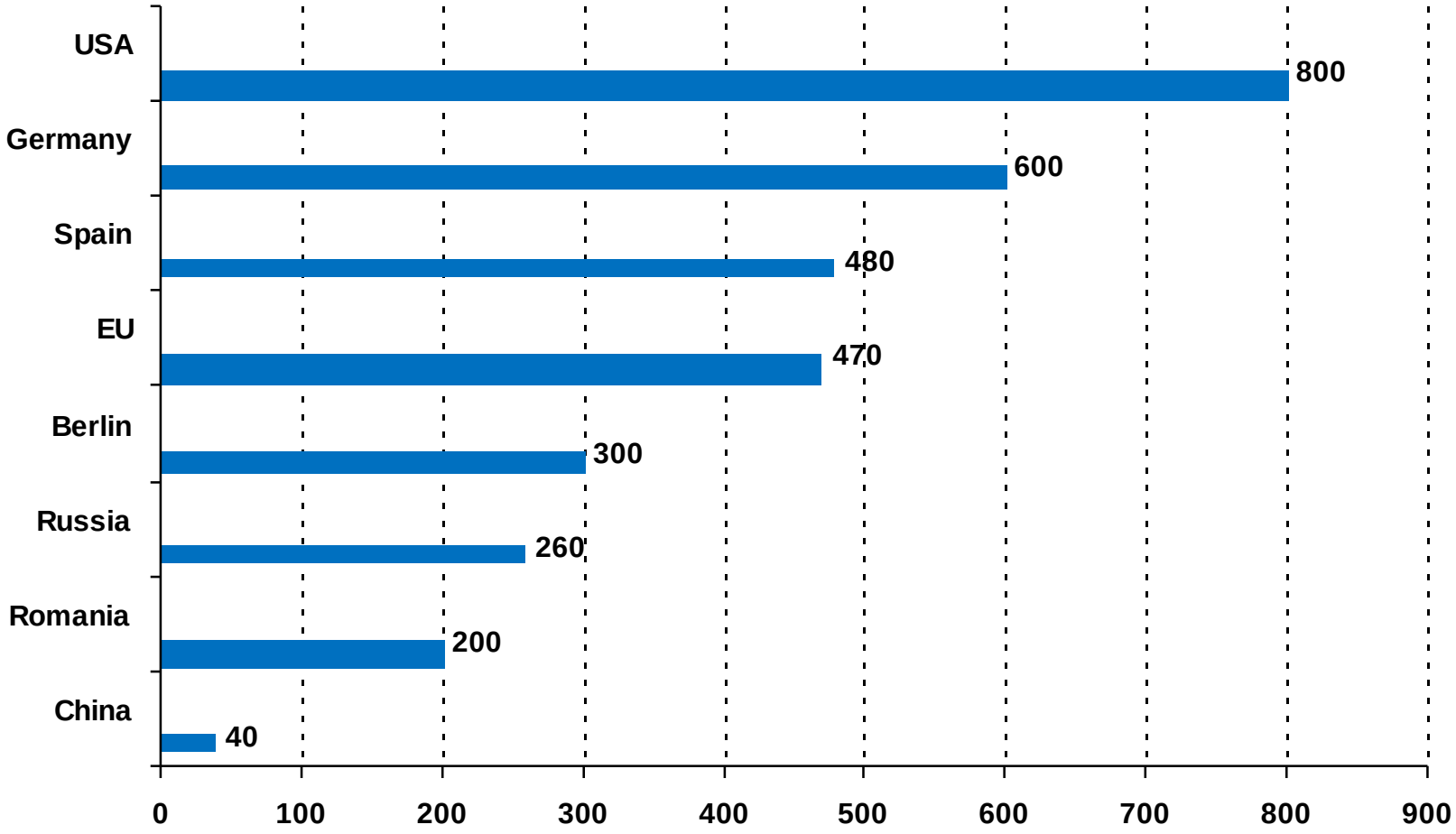
Victoria Transport Policy Institute, "Evaluating Transportation Land Use Impacts"

Automobile Dependency

Car dependency has increased significantly over the years

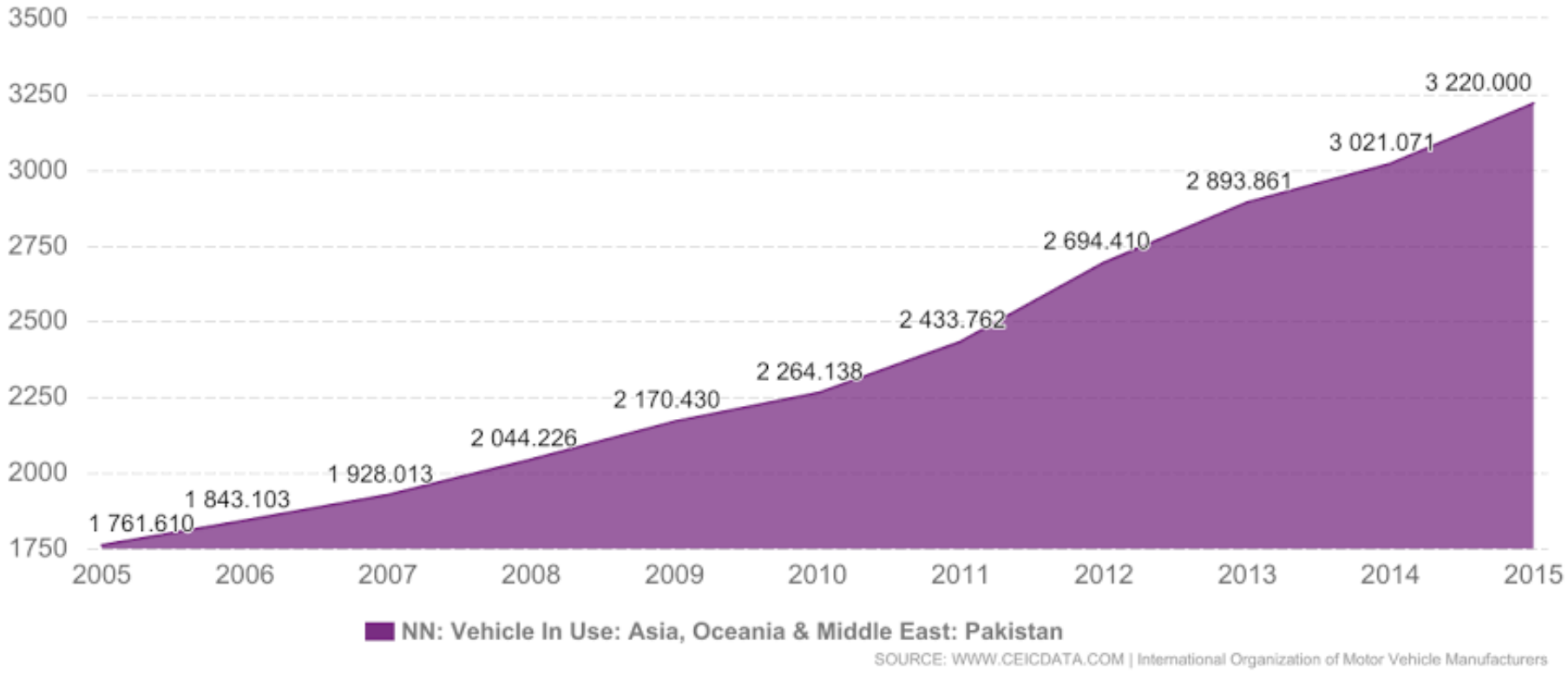


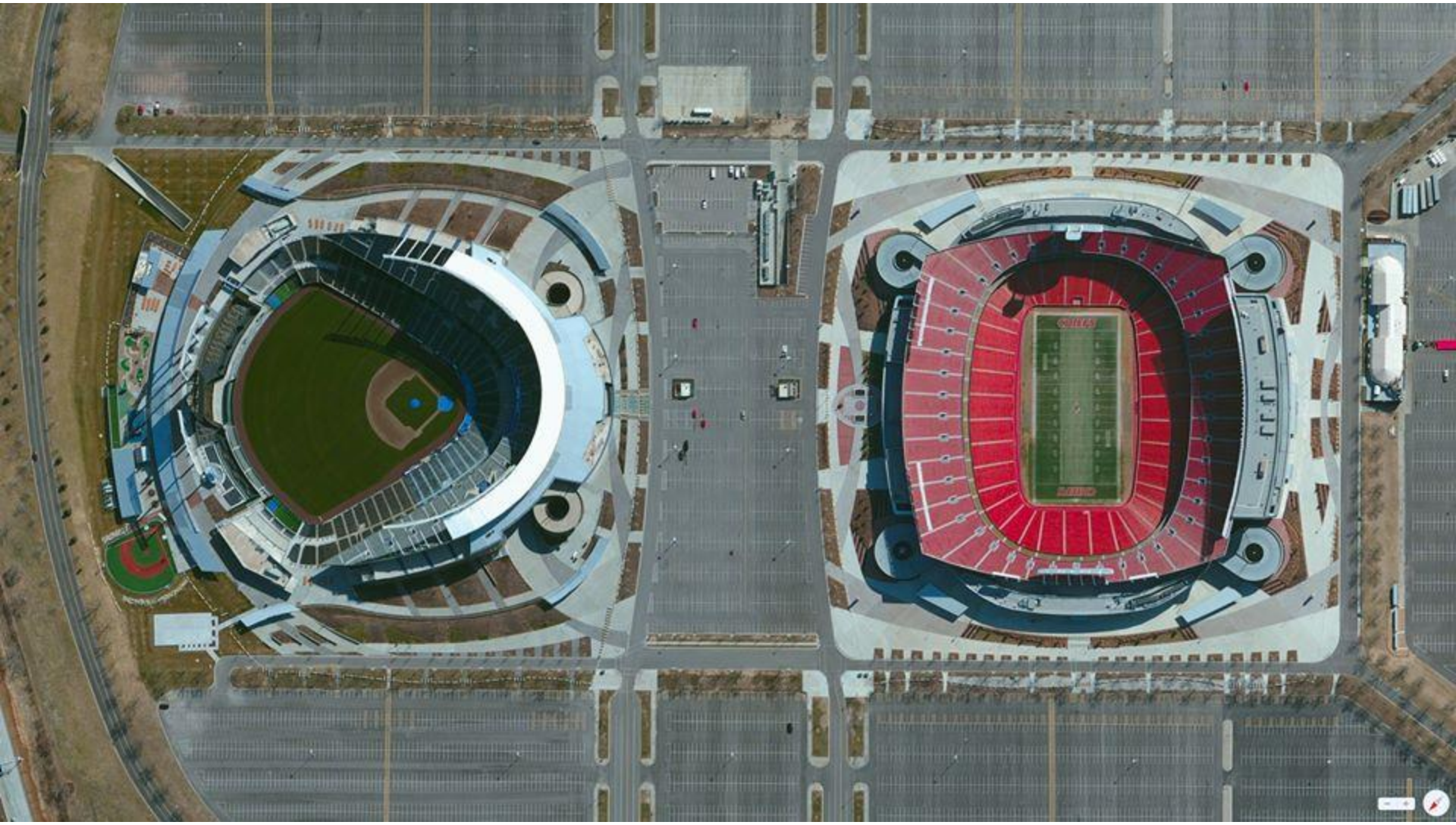
Cars per 1000 inhabitants



Automobile Dependency

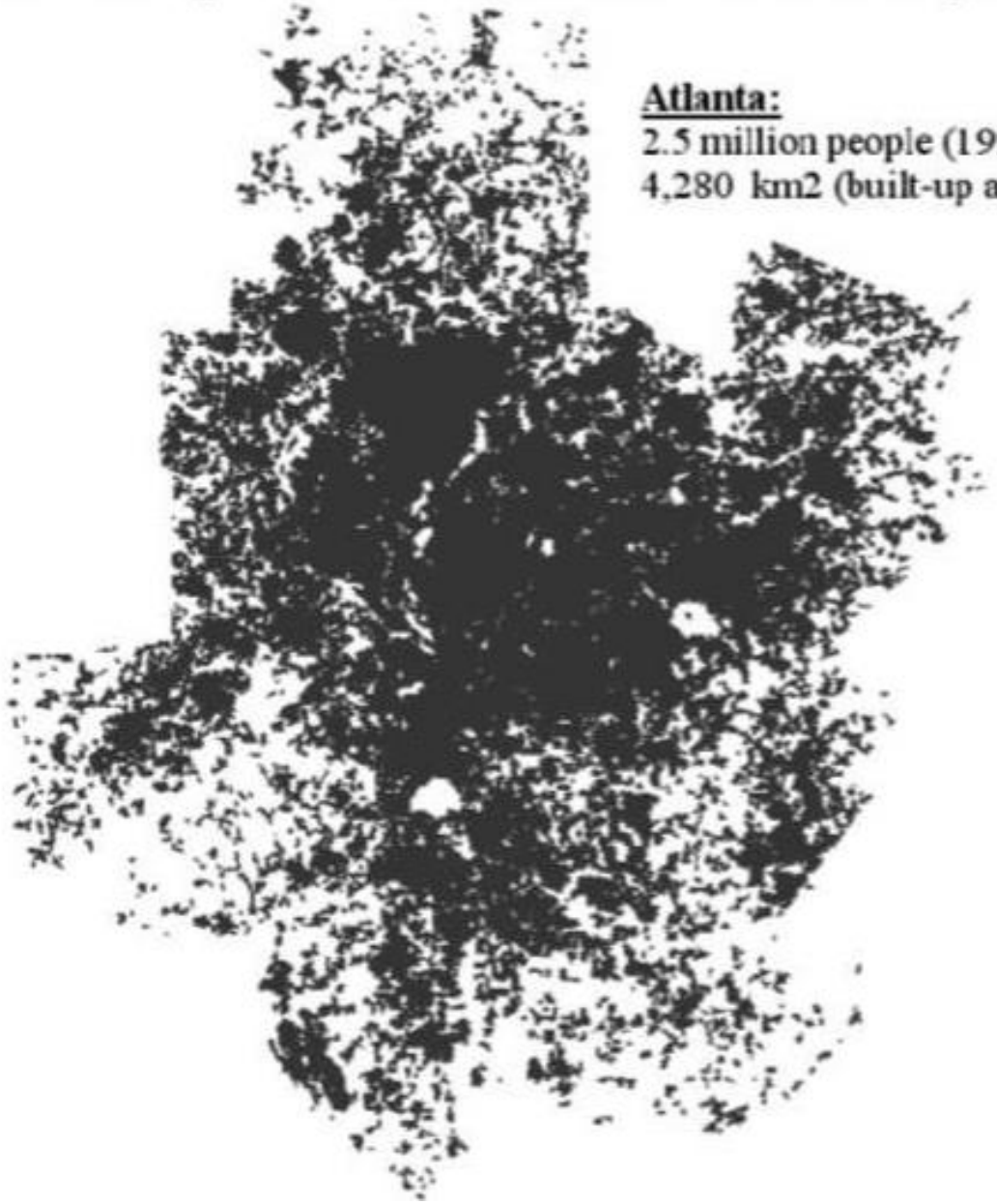
Pakistan Motor Vehicle Registered [2005 - 2015]





Arrowhead Stadium, Kansas City, Missouri, USA

The Built-up Area of Atlanta and Barcelona Represented at the Same Scale

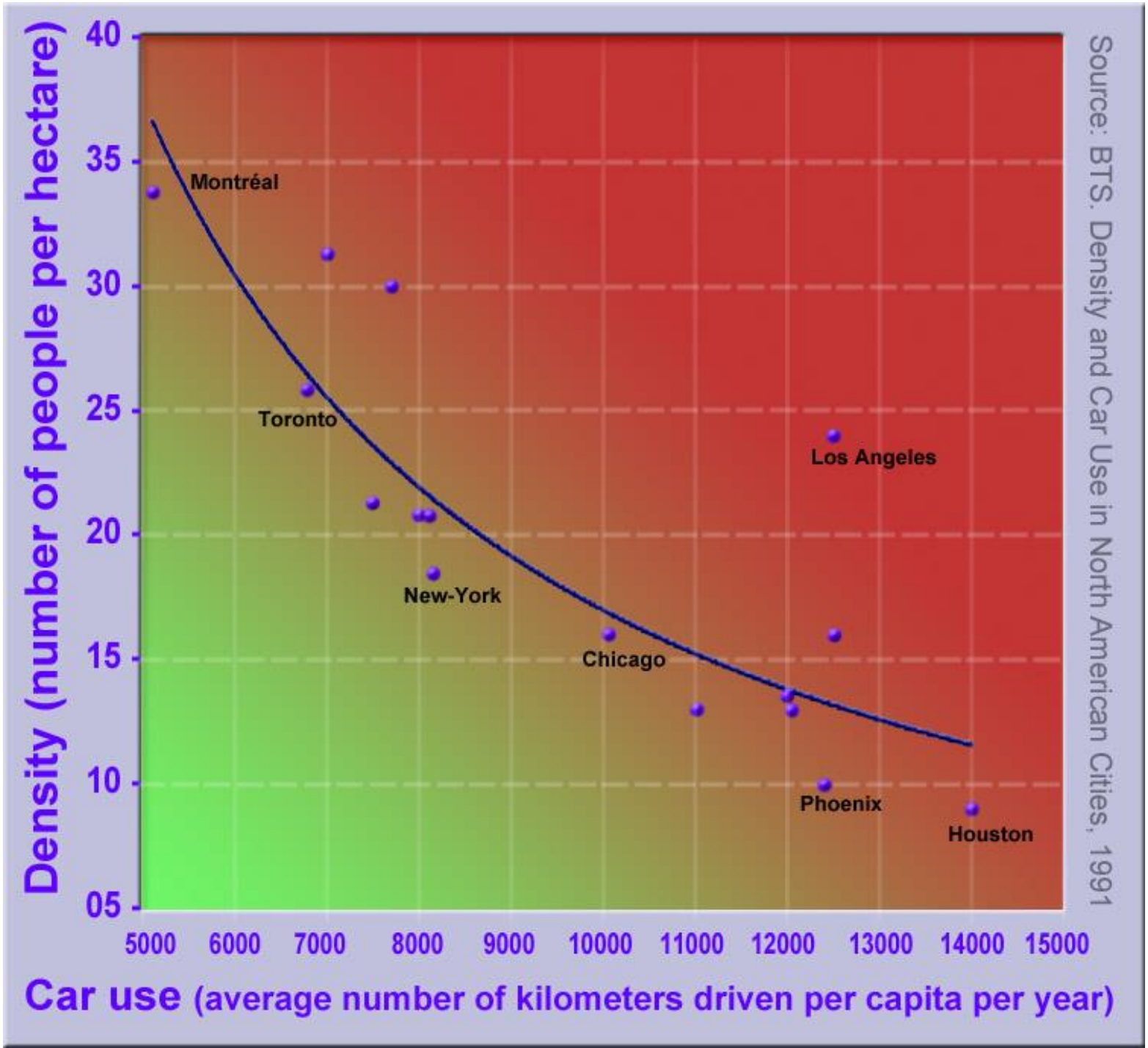


Atlanta:
2.5 million people (1990)
4,280 km2 (built-up area)

Barcelona:
2.8 million people (1990)
162 km2 (built-up area)



Source: Alain Bertaud, 2002


Density (number of people per hectare)
40
35
30
25
20
15
10
05
Montréal
Toronto
New-York
Chicago
Los Angeles
Phoenix
Houston
5000
6000
7000
8000
9000
10000
11000
12000
13000
14000
15000
Car use (average number of kilometers driven per capita per year)
Source: BTS. Density and Car Use in North American Cities, 1991


THANK
YOU!